

Dmitri Volkov

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Ridgefield CT, USA

EDUCATION

Indiana University, Bloomington IN, USA

2025 - present

M.S. in Computer Science

Research in weighted relational/logic programming language theory and compilation. GPA: 3.9.

Indiana University, Bloomington IN, USA

2021 - 2025

B.S. in Computer Science and B.M. in Music Composition

Highest distinction and honors. Minors in Computer Music and Mathematics, Computer Science concentrations in Programming Languages and Foundations. GPA: 3.99. Credits taken: 219.

Institute for Research and Coordination in Acoustics/Music (IRCAM), Paris, France

2024

Contemporary Music Creation and Critique course in Max/MSP and other IRCAM software.

EXPERIENCE

Internship at Old Road Software, Ridgefield CT, USA

2025

Full-stack development across iOS/Android/web/backend APIs, including OpenAI API integration.

Projects: legaltech form generator, nutrition supplement counterfeit scanner, AI wellness app.

Undergraduate Instructor at Indiana University, Bloomington IN, USA

2023 - present

Courses: CSCI-C311/H11/A595 Programming Languages, CSCI-A290/A590 Straight Line to Deep Learning. Review code, solidify student understanding, collaborate to develop course materials.

PROJECTS

***semiringKanren*, logic/relational programming language**

2025 - present

Typed polymorphic logic language computing over semirings. Can outperform faster-miniKanren, supports disequality constraints. Presented at ICFP 2025. github.com/sporkl/semiringkanren

***Pivotuner*, commercial audio plugin for macOS and Windows**

2021 - present

Virtual instrument pure intonation autotune. Developed with Grammy-winning/nominated musicians e.g. Jacob Collier, Nick Revel. \$2500+ in sales. Presented at MOXsonic 2024, SPLICE VI.

Hardware-accelerated matrix-vector multiplier (course project)

2025

32-bit float matrix-vector multiplier for Pynq-Z1, implemented in Verilog. Uses pipelined float FMAC IPs, supports arbitrary parallelization (up to hardware) through parametrization, beats NumPy.

Formally-verified finger tree (course project)

2025

Finger tree data structure verified in Rocq, including property-based tests and proofs of: operations preserving invariants, algebraic properties, and abstraction to lists. tinyurl.com/yc3cwfbf

Lisp-like language compiler (course project)

2024

Compiler from Lisp-like language to x86-64 assembly. Features: garbage collection, tail call optimization, compile-time evaluation, support for multithreaded programs. tinyurl.com/286t9248

Malt Transformer, transformer neural network implementation

2023

Library for GPT architecture as used by LLM AI models. github.com/sporkl/malt-transformer

PUBLICATIONS / PRESENTATIONS

Committing to the bit: Relational programming with semiring arrays and SAT solving

2025

Introduces semiringKanren logic language including denotational semantics, correctness-preserving SAT compilation, and applications. Presented at ICFP 2025. doi.org/10.48550/arXiv.2509.22614

Sonifications of Quantum Superpositions: Methods and Musical Applications

2025

First chapter of Advances in Quantum Computer Music, published by World Scientific. Illustrates general approach to quantum sonification, encapsulating most existing work. doi.org/10.1142/14025

Sonifying Quantum Superpositions by Correlating State Probability to Rhythmic Density

2023

Psychoacoustics-informed quantum sonification method, including musical and pedagogical considerations. Presented at ISQCMC 2023. doi.org/10.5281/zenodo.10206655

AWARDS / HONORS / SCHOLARSHIPS

Luddy Dean's Advisory Council Senior Scholarship (Indiana University Bloomington)	2024 - 2025
Luddy School of Informatics, Computing, and Engineering Dean's List	2023 - 2025
Sarah Brown Sherman Scholarship (Indiana University Bloomington)	2023 - 2025
National Merit Scholarship (Indiana University Bloomington)	2021 - 2025
Martin Eliot Grey Scholarship (Indiana University Bloomington)	2021 - 2025
Music Faculty Award Scholarship (Indiana University Bloomington)	2021 - 2025
Sigma Alpha Iota Inter-American Music Awards Winner	2025
RIOCC Organ Composition Competition Young Artist Division 1st Prize	2021

SKILLS

Programming Languages: C/C++, Python, TypeScript/JavaScript, (System)Verilog, OCaml, Rocq, SQL, Lua, R, PHP, Racket, Mathematica, Max/MSP/Pure Data, LaTeX, HTML/CSS, C#, Kotlin, Java, x86.

Frameworks/Libraries: JUCE, UNIX libc, Arduino, React Native/Expo, LOVE2d, Android(X), Vivado, Qiskit, NumPy, Pandas, matplotlib, scikit-learn, Azure Functions, Snowflake, PostgreSQL.

Specializations: compilers, formal methods/verification, FPGAs/RTL design/HDL/embedded, math, realtime digital audio/music, functional and logic/relational programming, quantum computing.

Miscellaneous: UNIX shell, technical writing, research, devops, Agile, Vim, VSCode, JetBrains IDEs, Xcode, macOS, Linux, Windows, debugging, optimizing, CMake, Git, web hosting, Microsoft Azure, APIs, Spanish, technical communication, music (composition/performance/production).